

SOFTWARE REQUIREMENTS SPECIFICATION

Hill Express — Grocery Quick-Delivery Platform

Version 1.0 · July 2026

Project	Hill Express — Grocery Quick-Delivery Platform
Version / Date / Status	1.0 · July 2026 · Draft for client review & sign-off
Prepared for	Hill Express
Prepared by / Approved by	_____ / _____

1. Introduction

1.1 Purpose & Background

This Software Requirements Specification (SRS) defines the requirements for Hill Express, a hyperlocal grocery quick-delivery platform, and is the baseline for development and client sign-off.

An earlier, comprehensive specification described a large multi-category quick-commerce system. To launch quickly within a fixed timeline, Hill Express and the development team have agreed to a focused initial scope built around one reliable loop: a customer orders groceries, the store confirms and packs the order, a driver is assigned, and the customer sees the driver's name, phone number, and expected time of delivery until delivery. This document supersedes the earlier one for the first release; features not listed here are deferred to later phases and handled via change request.

1.2 Scope

The software delivers the following. Features not listed here are addressed in later phases and handled via change request.

- Customer App (Android): browse a grocery store, cart, COD/online payment, place & track orders (driver name, number, ETA).
- POS / Store Console (Web): accept/reject orders, packing time, ready-for-pickup, inventory (manual + bulk sheet import).
- Driver App (Android): receive assigned orders, maps navigation, status updates, delivery OTP, COD.
- Admin Panel (Web): orders, driver assignment, stores/drivers/users, refunds, basic reports.
- Backend: REST APIs, relational DB, image storage, Firebase auth + push, single VPS.

1.3 Constraints & Assumptions

- Timeline ~2–3 months (8–12 weeks), driving a lean, Android-first, manual-dispatch build.
- Platforms: Android (Customer, Driver) + web (POS, Admin); no iOS at launch. Single city/service area; limited stores and drivers.
- Payments: COD + one online gateway. Navigation via the installed maps app (no paid map SDK).

Working assumptions to confirm before sign-off (changes may affect scope/timeline/cost):

- Multiple stores supported, each with its own POS for orders & inventory; may launch with a single store.
- Driver assignment is central/manual by Admin; the POS focuses on orders and stock.
- Driver side is a lightweight Android app (needed for push assignments).
- Customer notifications: in-app push + SMS for OTP only at launch (order-step SMS/WhatsApp deferred).
- ETA = store packing time + a fixed delivery buffer (or a flat default).
- One payment gateway (Razorpay / Cashfree / PhonePe) to be finalised; design stays gateway-agnostic.

1.4 Definitions & Abbreviations

Term	Meaning
POS	Point of Sale — the store console used to accept orders and manage inventory.
COD	Cash on Delivery.
OTP	One-Time Password (SMS code) for login and delivery/pickup verification.

Term	Meaning
SKU	Stock-Keeping Unit — a unique code identifying a product in inventory.
ETA	Estimated Time of Delivery shown to the customer.
FCM	Firebase Cloud Messaging — the push-notification service.

2. Overall Description

2.1 Applications & Roles

Application / Role	Description
Customer App (Android)	Customers browse the store, place & pay for orders, track delivery, and optionally rate.
POS / Store Console (Web)	Store staff accept/reject orders, set packing time, mark ready, verify pickup, and manage inventory.
Driver App (Android)	Drivers receive assigned orders, navigate, update status, verify delivery OTP, and handle COD.
Admin Panel (Web)	Admin assigns drivers, manages stores/drivers/users/orders, processes refunds, and exports reports.

2.2 Operating Environment

- Customer App and Driver App: Android smartphones.
- POS and Admin Panel: modern web browsers on desktop/laptop.
- Backend: a single VPS/managed host with a relational database, object storage for images, and Firebase for auth & push.

2.3 System Modules

User Management · Address & Serviceability · Store Management · Product/Inventory (manual + bulk import) · Cart & Checkout · Order Management · Driver Management · Manual Dispatch · Order Tracking (status + driver + ETA) · Payment (COD + gateway) · COD Handling · Notifications (push + OTP SMS) · Ratings (optional) · Admin Roles (limited) · Reporting & Export.

2.4 Core Order Flow

1. Customer selects a serviceable address, browses the store, adds items, and checks out.
2. Customer chooses COD or online payment and places the order (unique Order ID generated).
3. The order appears on the store POS; the operator accepts it and sets a packing time.
4. Admin assigns a driver (manual dispatch); the customer sees driver name, number, and ETA.
5. Driver reaches the store, verifies the pickup OTP/code, and marks picked up.
6. Driver delivers, verifies the delivery OTP, collects COD if applicable, and marks delivered; payment settles and the order closes.

3. Functional Requirements — Customer App

ID	Requirement
FR-C-001	Register and log in using mobile number + OTP; prevent duplicate accounts for the same number (email/Google login optional, not required in this release).
FR-C-002	Create/update the user profile; account status is Active or Blocked. Profile fields: • Name • Mobile number • Email (optional) • Profile photo (optional)

ID	Requirement
	Default address
FR-C-003	Address: detect via GPS or enter manually, and save up to two (e.g., Home/Work). Each address captures: - Address label - House / building name - Street / locality - Landmark - City / town - Pin code - Latitude & longitude - Delivery instructions
FR-C-004	Serviceability: validate the address against a serviceable zone (pincode or radius) and block ordering with a clear message when outside the area.
FR-C-005	The home screen shall display: - Selected delivery address - Search bar - Grocery categories - Available store(s) with open/closed status - Active-order tracking card - Offers / banner (optional) - Estimated delivery time
FR-C-006	Category browsing (adjustable): - Staples & Grains - Fruits & Vegetables - Dairy & Eggs - Snacks & Packaged Foods - Beverages - Personal Care - Household & Cleaning - Bakery - Baby & Others
FR-C-007	Search by product name, category, or brand; filter by category / in-stock / price / offers; sort by relevance, price, or popularity.
FR-C-008	Product listing shall show (out-of-stock items are marked/hidden and cannot be added): - Name - Image (optional) - Unit / pack size (kg / g / litre / ml / piece / pack) - Price & discounted price - Availability / stock status - Variants (e.g., 500 g / 1 kg)
FR-C-009	Cart: add / update quantity / remove; re-validate availability & price at checkout; block checkout if the store is closed or the address is unserviceable. The cart shall show: - Item subtotal - Delivery charge - Discount / coupon (if any) - Tax (if applicable) - Final payable amount
FR-C-010	Place a grocery order (unique Order ID) and maintain a status timeline. Order cancellation is supported based on the order stage (e.g., before the store accepts).
FR-C-011	Payment: choose COD or online at checkout; retry on failure; store a transaction reference for online

ID	Requirement
	payments; payment status is Pending, Paid, Failed, Refunded, or Partially Refunded. Modes: • Cash on Delivery • UPI • Debit / credit card • Net banking (as supported by the gateway)
FR-C-012	[CORE] Track order status through the defined steps; after a driver is assigned, show the driver's name, phone number, vehicle type (if available), and ETA, with one-tap (masked where enabled) calling. Live map tracking is out of scope for this release.
FR-C-013	Notifications: push for order placed / accepted / packing / driver-assigned / out-for-delivery / delivered / cancelled and payment & refund events; OTP via SMS.
FR-C-014	Order history with items, amount, status, and date; optional quick reorder; optional 1–5 star rating after delivery; "Contact Support" via phone/WhatsApp (no in-app ticketing in this release).

4. Functional Requirements — POS / Store Console

ID	Requirement
FR-P-001	Store-operator login (secure credentials and/or OTP); a store open/close toggle controlling order intake; store profile shows name, address, contact, delivery fee (set by Admin), and hours.
FR-P-002	Receive new orders in real time, each showing: • Order ID • Items & quantities • Customer area / locality • Payment mode (COD / Paid) • Order total • Delivery instructions (if any)
FR-P-003	Accept or reject an order (rejection requires a reason); set/confirm the packing (ready) time on acceptance (feeds the customer ETA); mark "Packed / Ready for Pickup".
FR-P-004	Verify the assigned driver at pickup using a pickup OTP/code; view active and completed orders with basic filters (status, date).
FR-P-005	[Inventory — manual] Add a product; edit, disable, delete, and mark available/out-of-stock. Product fields: • Name • Category • Unit (kg / g / litre / ml / piece / pack) • Pack size / variant • Price • Discount (optional) • Tax (optional) • Stock quantity • Image (optional) • Availability
FR-P-006	[Inventory — bulk import] Add/update many products at once via a sheet: • Download a CSV/Excel template (columns in Section 7) • Upload a filled sheet to add or update many products • System validates the file and reports rows added, updated, and failed (with reasons) • Re-import updates price, stock, and availability matched by SKU/code (or name)
FR-P-007	Low-stock indicator (optional). On stock-out after an order, the operator can reject the order or flag items for Admin to refund/adjust. Partial fulfilment is a later-phase item.

5. Functional Requirements — Driver App

ID	Requirement
FR-D-001	Driver login via mobile OTP; set status Online / Offline / On Pickup / On Delivery / On Break; capture location while online (for Admin & assignment; no customer-facing live map).
FR-D-002	Receive a manually assigned order via push (accept the task; default admin-assigned), showing: • Store name & address • Customer name / number / address • Items summary • Payment mode • COD amount (if applicable)
FR-D-003	[Workflow] Follow the delivery workflow; "Delivered" cannot be marked without the delivery OTP unless Admin overrides, and each update reflects to the customer and Admin: • Reached pickup • Verify pickup OTP / code • Picked up • Out for delivery • Reached customer • Verify delivery OTP • Collect COD (if any) • Delivered
FR-D-004	Navigation opens the installed maps app (e.g., Google Maps) for pickup and drop, showing addresses and customer contact.
FR-D-005	Show COD to collect per order and a simple daily summary (deliveries completed and COD collected); optional cash-deposit-due. Earnings/incentive computation and rider analytics are out of scope.

6. Functional Requirements — Admin Panel

ID	Requirement
FR-A-001	Secure admin login; roles: Admin (full access) and Operations/Support (limited). Full multi-role RBAC is deferred.
FR-A-002	Dashboard operational counts (no BI charts): • Orders today • Active orders • Delivered orders • Cancelled orders • COD to be collected / collected
FR-A-003	Order management: • View all; search by Order ID; filter by status/date/store • View order details & status timeline • Assign / reassign a driver (manual dispatch) • Cancel an order and record the reason • Approve refunds (online) or COD adjustments
FR-A-004	Stores: add/edit; set delivery fee and hours; open/close; oversee catalogue (enable/disable products); block/unblock.
FR-A-005	Drivers: add/edit (manual onboarding); active/inactive; view status, current assignment, and location (basic); set a per-driver COD limit (optional).
FR-A-006	Users: view/search; view order history; block/unblock; make a refund/adjustment where applicable.

ID	Requirement
FR-A-007	Reports: export CSV/Excel (broader analytics deferred): - Daily Orders report - COD Collection report - Store Sales summary

7. Payments, Notifications, Data & Import

7.1 Payments

COD: the order is confirmed without upfront payment; the driver collects the exact amount at delivery, the order is marked paid (COD collected), and it appears in the driver summary and the COD report. Online: the customer pays via the gateway (UPI/cards/net-banking); on success the order is marked Paid with a stored transaction reference, and the gateway webhook/callback updates status reliably; on failure the customer can retry or switch to COD. Refunds: initiated by Admin per cancellation stage/complaint outcome — online refunds go to the original method via the gateway; uncollected COD needs no monetary refund. Gateway fees apply per online payment.

7.2 Notifications

Customer and Driver receive push notifications; the POS and Admin consoles reflect the same events; SMS is used for OTP.

Event	Customer	Driver	Store/POS	Admin
Order placed	Yes	No	Yes	Yes
Order accepted / rejected	Yes	No	Yes	Yes
Packing / ready for pickup	Yes	Yes	Yes	No
Driver assigned	Yes	Yes	Yes	Yes
Picked up	Yes	Yes	Yes	Yes
Out for delivery	Yes	Yes	No	No
Delivered	Yes	Yes	Yes	Yes
Cancelled	Yes	Yes	Yes	Yes
Payment success / failed	Yes	No	No	Yes
Refund processed	Yes	No	No	Yes
OTP (login / delivery)	SMS	SMS	—	—

7.3 Core Data

Entities: Users, User Addresses, Stores, Categories, Products/Items, Cart, Orders, Order Items, Order Status History, Payments, Refunds, Drivers, Delivery Zones, Coupons (optional), Admin Users & Roles, Notifications.

Order fields: Order ID, User ID, Store ID, Driver ID, order status, payment status & mode, item total, delivery charge, discount, tax, final amount, COD amount, coupon ID (optional), placed/accepted/ready/assigned/picked-up/delivered times, ETA, cancellation reason, refund status. Product fields: Product ID/SKU, Store ID, name, category, unit, pack size/variant, price, discount, tax (optional), stock quantity, availability, image URL, updated date.

7.4 Inventory Import Template (Columns)

Column	Notes
SKU / Product Code	Optional but recommended; used to match rows on re-import.
Product Name / Category	Required; category must match an existing one.
Unit / Pack Size	Unit: kg/g/litre/ml/piece/pack; pack size e.g. 500 g, 1 kg, 6-pack.

Column	Notes
Price / Discount	Price required (numeric); discount optional.
Stock Quantity / Availability	Stock required (numeric); availability Yes/No.
Image URL	Optional.